

1. Project Context

Starlight Data Solutions is pleased to present this Technical Proposal to the <CUSTOMER-NAME>, for the deployment of an AI-Powered Intelligent Video Surveillance System across the <CUSTOMER-NAME>.

The proposed solution leverages state-of-the-art Artificial Intelligence, Computer Vision, and Machine Learning technologies to transform the existing CCTV infrastructure into an active, real-time intelligence platform. This system will empower the <CUSTOMER-NAME> Police to proactively detect, alert, and respond to crimes, traffic violations, public safety threats, and narcotics-linked activities — significantly improving law and order across the district.

The proposal is structured across three distinct deployment phases, each aligned with the operational priorities of the <CUSTOMER-NAME> police:

- Phase 1 – Non-Negotiable (Urgent): Immediate deployment of critical use cases covering stolen vehicle detection, case-linked vehicle alerts, and face recognition for wanted persons.
- Phase 2 – Must Have: Deployment of 14 high-priority use cases covering crime detection, traffic enforcement, crowd management, and system integrity monitoring.
- Phase 3 – Good to Have: Advanced deployment of 18 use cases covering predictive analytics, narcotics surveillance, gender safety, fire detection, and intelligent reporting.

Installed Camera Details:

District Police has these cameras as discussed:

Phulbani Town CCTV – Camera Details

Total New Cameras: 76

- AI Cameras: 49
- Non-AI Cameras: 27

AI Cameras

Sl. No.	Brand	Model	Type	Quantity
1	Dahua	DH-IPC-HFW4841T-ZAS	Bullet Camera	15 pcs
2	Dahua	DH-IPC-HFW3841TP-ZAS	Bullet Camera	5 pcs
3	Hikvision	ANPR	ANPR Camera	6 pcs
4	Hikvision	DS-2CCD3646G2T-IZS	Motorized Camera	23 pcs

Non-AI Cameras

Sl. No.	Brand	Type	Quantity
1	Hikvision	PTZ	2 pcs
2	-	Bullet	25 pcs

4. Phase 1 – Non-Negotiable (Urgent) Use Cases

Phase 1 covers three critical use cases that are mandatory and urgent for <CUSTOMER-NAME> Police. These use cases form the foundation of the AI surveillance system and deliver immediate operational value from the day of deployment. The focus areas are vehicle intelligence and wanted persons detection.

Phase 1 – Priority: Non-Negotiable (Urgent) | Use Cases: 3

#	Category	Use Case	Solution Capability
1	Vehicle	Stolen Vehicle Detection (OD Registered)	Reads ANPR from all OD-registered plates across CCTVs. Matches each plate against the stolen vehicle database in real time. Supports API integration with state-level and <CUSTOMER-NAME> district stolen vehicle databases. Allows district police to update records, search vehicle numbers, track movement across CCTVs, generate intelligence reports, and receive real-time stolen vehicle alerts.
2	Vehicle	Case-Linked Vehicle Alert (OD Registered)	Matches all OD-registered plates against the police case management database (FIRs, warrants, watchlists). Any match triggers an instant alert with vehicle details, associated case number, and location to the concerned officers.
3	Crime	Face Recognition – Wanted Persons	AI facial recognition integrated with the police wanted persons database. When a match is found in any CCTV feed, an alert with face image, camera location, and matching criminal profile is instantly sent to relevant officers. Supports both manual and automated alerting.

4.1 Phase 1 – Implementation Highlights

- Mandatory API integration with the <CUSTOMER-NAME> district and Odisha state stolen vehicle databases
- District police will have full control to update stolen vehicle records in real time
- Complete vehicle movement tracking across all CCTV cameras with intelligence report generation
- Facial recognition engine will be integrated with the current police wanted persons database
- Both manual and automated alert triggering supported for face-matched events
- Instant alerts delivered to concerned officers with face image, location, and criminal profile

5. Phase 2 – Must Have Use Cases

Phase 2 comprises 14 high-priority use cases that are essential for comprehensive crime prevention, traffic management, and system reliability across the district. This phase transforms the CCTV network into a proactive policing tool, enabling faster response times and automated penalty generation.

Phase 2 – Priority: Must Have | Use Cases: 14

#	Category	Use Case	Solution Capability
1	Crime	Fight / Physical Assault Detection	AI human pose estimation and motion analysis detects abnormal body postures associated with fighting — rapid limb movements, falls, grouped aggression. Triggers real-time alert with camera location and live feed link for immediate police response.
2	Crime	Chain Snatching / Robbery Pattern Detection	AI model analyses proximity of two-wheelers to pedestrians, sudden approach patterns, and post-event rapid departure. On matching a robbery signature, an alert is raised immediately with footage clip for the nearest patrol unit.
3	Crime	ATM Vandalism and Theft Detection	Deployed on CCTV near ATMs. Detects tampering behaviour — excessive touching of panels, persons beyond a time threshold in off-hours, or tool-like objects near the machine. Raises a priority alert to the police data center instantly.
4	Crime	Missing Person / Child Detection	Police upload the missing person's photo to the system. The AI facial recognition model immediately scans all live and past CCTV feeds and alerts officers the moment a match is found with camera location and timestamp.
5	Crowd	Crowd Gathering Detection and Alert	AI crowd density estimation model continuously monitors all CCTV feeds. When the number of people exceeds a defined threshold or abnormal crowd movement is detected, an instant notification is sent to senior officers with live feed access.
6	System	CCTV Tampering / Camera Obstruction Detection	Monitors video feed quality of each camera continuously. Any sudden loss of feed, blurring, physical blockage, or tampering is detected instantly. Alerts are raised to the technical team and police officers with camera ID and location.
7	Traffic	Helmet Violation Detection	Detects two-wheeler riders without helmets in real time. Captures rider image and vehicle plate. Auto-generates a penalty challan with timestamp, location, and vehicle number in the police portal. Optional API integration with RTO.
8	Traffic	Seat Belt Violation Detection	Detects absence of seat belt in car drivers and front passengers. Captures image, logs violation with location and plate number, and creates a penalty entry automatically. Optional RTO API integration.
9	Traffic	Mobile Phone Use While Driving Detection	Detects drivers holding or looking at a mobile phone while driving. Captures image evidence with timestamp, plate number, and location. Generates a penalty record in the police portal. Optional RTO API integration.
10	Traffic	Over speeding Vehicle Detection	Uses frame-to-frame vehicle tracking across CCTVs to calculate speed. If speed exceeds the defined limit, captures vehicle image, plate, calculated speed, location, and time, and raises a penalty alert. Optional RTO integration.
11	Traffic	Red Light / Signal Jumping Detection	Syncs with traffic signal status and detects vehicles crossing the stop line during a red signal. Captures vehicle image with plate, time,

#	Category	Use Case	Solution Capability
			and location, and creates a challan entry in the penalty portal. Optional RTO API integration.
12	Traffic	Triple Riding on Two-Wheeler Detection	Counts the number of persons on a two-wheeler in real time. If three or more persons are detected, captures image, records plate number, time, and location, and creates a penalty entry automatically. Optional RTO API integration.
13	Vehicle	Wrong Way / One-Way Violation Detection	Detects vehicles moving against designated traffic direction in one-way streets. Captures vehicle image, plate, direction, and location, and sends a penalty alert and live notification to the nearest beat officer.
14	Vehicle	Overloaded / Modified Vehicle Detection	Estimates passenger count and cargo load from CCTV feeds. Flags vehicles exceeding capacity limits or showing illegal structural modifications with a portal alert, image evidence, and location for RTO/police action.

5.1 Phase 2 – Implementation Highlights

Crime Detection

- AI models for fight detection, robbery pattern analysis, and missing person tracking provide 24x7 active crime monitoring
- ATM vandalism detection is deployed specifically at ATM locations with immediate priority alerts

Traffic Enforcement

- Six traffic violation types auto-detected with system-generated challans — helmet, seat belt, mobile usage, overspeeding, red light jumping, and triple riding
- Optional API integration with the RTO management console enables automated challan processing
- Wrong-way and overloaded vehicle detection reduces road accidents and illegal transport

Crowd and System Management

- Crowd density monitoring enables timely mobilization of police during public events and gatherings
- CCTV tampering detection ensures the integrity of the surveillance network at all times

6. Phase 3 – Good to Have Use Cases

Phase 3 represents the advanced layer of the AI surveillance platform, introducing predictive analytics, narcotics intelligence, gender safety mechanisms, fire detection, and specialized crime monitoring. These 18 use cases elevate <CUSTOMER-NAME> Police to a smart, data-driven policing organization capable of anticipating threats before they escalate.

Phase 3 – Priority: Good to Have | Use Cases: 18

#	Category	Use Case	Solution Capability
1	Analytics	Crime Hotspot Heatmap and Predictive Patrol Deployment	Analyses historical CCTV alert data, FIR records, and incident reports to generate crime hotspot heatmaps by location and time. Suggests optimized patrol deployment schedules to SP-level officers through the analytics dashboard.

#	Category	Use Case	Solution Capability
2	Analytics	Daily AI-Generated Police Intelligence Report	Auto-generates a daily intelligence summary — alerts raised, top crime-prone zones, frequent violators, active wanted persons, narcotics vehicle patterns — delivered to the SP, DSP, and district control room every morning at a defined time.
3	Crime	Abandoned Object / Suspicious Package Detection	Monitors stationary objects over time. If an object remains unattended beyond a defined threshold in a sensitive area, an alert with image, location, and time is sent immediately to bomb squad and local police.
4	Crime	Night-Time Loitering Detection	Detects persons loitering in geo-fenced sensitive zones beyond defined night hours (e.g., 11 PM to 5 AM). Night-vision compatible model raises alerts with captured image and location to the on-duty officer.
5	Crime	Weapon Detection in Public Areas	AI object detection model trained on weapon categories (knife, rod, firearm, sharp objects) detects such items in real time from CCTV. Triggers an emergency alert to the nearest patrol unit with feed location and screenshot.
6	Crime	Road Accident Detection and Emergency Alert	Detects sudden vehicle collisions, overturns, or pedestrian knockdown events. Triggers a high-priority alert simultaneously to traffic police, local police station, and emergency medical services with live feed access.
7	Crime	Pickpocketing Pattern Detection	Detects suspicious hand movements near other persons' pockets or bags in dense crowd settings. Rapid contact-and-retreat patterns trigger an alert with time, location, and captured image to the nearest plainclothes officer.
8	Crime	Juvenile / Minor Driving Detection	Estimates the driver's age using facial analysis. If the driver appears below the legal driving age (18 years), captures vehicle image, plate, and driver face, and raises an alert to traffic police for interception.
9	Fire / Safety	Fire and Smoke Detection in Public Areas	Analyses CCTV frames in real time for visual indicators of fire or smoke. Immediately notifies the fire department, local police, and district emergency control room with camera location, time, and live feed link.
10	Gender Safety	Eve Teasing / Public Harassment Detection	Detects patterns linked to public harassment — persistent following, aggressive approach, surrounding a lone individual. Triggers a priority alert to the women safety unit and nearest beat officer when such patterns are detected.
11	Narcotics	Suspicious Vehicle Movement Pattern Detection	Trained using police human intelligence (routes, timings, hotspots). Monitors vehicle movement patterns in CCTV and raises an alert if a vehicle matches a narcotics-linked movement signature.
12	Narcotics	Repeated Vehicle Rendezvous Detection	Identifies vehicles that meet repeatedly at the same location within specific time windows. Cross-references with known narcotics hotspots and raises an alert for the anti-narcotics team with vehicle details and footage.
13	Narcotics	Street-Level Drug Transaction Behavior Detection	Identifies suspicious brief exchanges — short stop, hand contact, quick departure — in known narcotics hotspot zones. Cross-referenced with the police intelligence database, any match triggers an alert with clip and location.
14	Traffic	No Parking Zone Violation Detection	Uses geo-fenced zone mapping on CCTV feeds. Detects any vehicle parked in a designated no-parking zone automatically. Captures vehicle image, plate, timestamp, and location for penalty issuance. Optional RTO API integration.

#	Category	Use Case	Solution Capability
15	Traffic	Drunk Driving Behavior Detection	Detects erratic driving patterns — lane weaving, inconsistent speed, sudden braking, near-miss events. Flags vehicles with location and clip; alert sent to the nearest patrol to intercept. Optional RTO API integration.
16	Vehicle	Vehicle Color Tampering Detection	Compares vehicle color captured in CCTV against registered OEM color from the RTO database. If a mismatch is detected, an automatic alert is raised in the police portal with a timestamped image and vehicle location.
17	Vehicle	Out-of-State Vehicle Detection	AI-based ANPR reads all vehicle plates in CCTV footage. Any vehicle with a non-OD registration number triggers an immediate alert in the portal with location, time, and plate image.
18	Vehicle	Tampered or Obscured Number Plate Detection	Analyses clarity, alignment, and readability of every detected number plate. If a plate is blurred, covered, tilted, or shows signs of tampering, raises an alert with vehicle image, location, and camera ID for police action.

6.1 Phase 3 – Implementation Highlights

Predictive Analytics

- Crime hotspot heatmaps generated from historical FIR and alert data help optimize patrol deployment
- AI-generated daily intelligence reports delivered to the SP, DSP, and district control room at a defined time each morning

Narcotics Surveillance

- Three dedicated narcotics detection models — suspicious vehicle movement, repeated vehicle rendezvous, and street-level drug transaction detection — are trained using police intelligence inputs on known routes, hotspots, and timing patterns

Gender Safety

- Eve teasing and public harassment detection alerts the women safety unit and nearest beat officer in real time

Fire and Emergency Safety

- Fire and smoke detection notifies fire department, police, and district emergency control room simultaneously with live feed access

Advanced Crime Use Cases

- Abandoned object, weapon detection, road accident, juvenile driving, and pickpocketing detection models provide end-to-end coverage of high-risk public scenarios

Vehicle Intelligence

- Color tampering, out-of-state vehicle detection, and tampered number plate detection add another layer of vehicle intelligence to the existing ANPR system